

SALEKHARD, Russian Federation

WMO: 233300

Lat: 66.5319N

Lon: 66.6744E

Elev: 16

StdP: 101.13

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.5	-38.9	-43.1	0.1	-41.0	-40.9	0.1	-38.6	9.5	-21.3	8.3	-20.6	1.3	180	0.788

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	25.4	18.6	22.8	17.1	20.5	15.7	19.2	24.0	17.8	21.8	16.4	20.0	2.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.4	12.5	21.7	16.0	11.4	20.2	14.5	10.3	18.7	55.0	23.9	50.3	21.8	46.1	20.1	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.9	7.5	6.5	-42.6	28.3	3.7	2.0	-45.2	29.7	-47.4	30.9	-49.5	32.1	-52.2	33.5		
			-42.4	21.5	3.4	1.8	-44.8	22.8	-46.8	23.8	-48.7	24.9	-51.2	26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.3	-23.5	-22.7	-14.8	-7.8	-0.1	10.1	14.4	11.3	5.5	-3.0	-14.9	-20.0
	DBStd	14.88	9.00	8.98	7.95	6.88	5.67	5.17	4.32	3.64	3.79	5.95	8.77	9.33
	HDD10.0	5891	1038	915	768	533	318	62	8	27	144	402	748	929
	HDD18.3	8666	1297	1148	1026	783	572	251	137	219	386	661	998	1188
	CDD10.0	289	0	0	0	0	3	66	145	67	8	0	0	0
	CDD18.3	22	0	0	0	0	0	4	16	2	0	0	0	0
	CDH23.3	172	0	0	0	0	0	38	123	11	0	0	0	0
	CDH26.7	28	0	0	0	0	0	5	23	1	0	0	0	0
Wind	WSAvg	2.8	2.3	2.4	2.7	3.2	3.3	3.4	3.1	2.7	2.7	2.8	2.5	2.4
Precipitation	PrecAvg	446	22	19	21	26	36	55	64	62	47	43	28	25
	PrecMax	649	45	43	46	78	81	124	181	159	131	76	59	47
	PrecMin	276	7	3	0	2	10	12	6	18	7	6	8	1
	PrecStd	82	9	9	10	14	18	27	38	33	23	15	11	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.9	-1.1	2.6	7.1	19.0	27.1	29.1	24.7	18.7	10.5	2.0	-0.2
		MCWB	-2.8	-2.0	0.8	4.1	12.9	17.6	20.1	18.4	15.0	8.7	1.2	-1.3
	2%	DB	-4.8	-4.3	0.7	4.5	14.4	23.9	26.6	21.6	15.2	8.1	0.5	-2.7
		MCWB	-5.5	-5.0	-0.6	2.4	9.7	16.6	19.3	16.7	11.8	6.3	-0.1	-3.3
	5%	DB	-7.5	-6.7	-1.2	3.2	10.3	21.2	24.3	19.6	12.9	5.8	-0.9	-4.9
		MCWB	-8.0	-7.2	-2.3	1.6	7.0	15.1	18.5	15.4	10.5	4.6	-1.6	-5.4
	10%	DB	-10.2	-9.3	-3.0	2.0	7.3	18.7	21.7	17.7	11.2	4.1	-3.2	-7.3
		MCWB	-10.6	-9.8	-3.9	0.7	4.8	13.7	16.9	14.4	9.2	3.1	-3.8	-7.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.6	-2.0	1.0	4.6	13.1	18.9	22.0	19.7	15.2	9.1	1.2	-1.2
		MCDB	-2.0	-1.2	2.1	6.5	18.4	24.6	26.5	23.6	18.1	10.2	1.8	-0.5
	2%	WB	-5.4	-4.9	-0.6	2.7	9.8	17.3	20.3	17.3	12.5	6.4	0.1	-3.3
		MCDB	-4.8	-4.4	0.5	4.0	13.9	22.9	25.2	20.3	14.5	7.8	0.5	-2.7
	5%	WB	-8.0	-7.1	-2.2	1.7	7.4	15.8	18.8	16.0	10.9	4.6	-1.6	-5.4
		MCDB	-7.5	-6.7	-1.3	2.9	10.2	20.6	23.4	18.9	12.5	5.7	-1.0	-4.9
	10%	WB	-10.5	-9.9	-3.9	0.7	5.0	14.1	17.5	14.7	9.3	3.2	-3.8	-7.8
		MCDB	-10.1	-9.3	-3.0	1.9	7.1	17.9	21.3	17.3	11.1	4.1	-3.2	-7.4
Mean Daily Temperature Range	5% DB	MDBR	7.5	7.7	8.9	8.7	6.9	7.9	8.4	7.7	6.1	5.2	6.9	7.4
		MCDBR	9.3	8.4	8.2	7.2	10.9	11.8	11.4	11.0	9.0	5.6	6.7	8.1
		MCWBR	9.1	8.0	7.3	5.8	7.0	6.2	5.7	6.6	6.1	4.5	6.4	7.8
	5% WB	MCDBR	9.2	8.4	8.0	6.8	10.8	11.1	10.7	9.6	8.2	5.4	6.6	8.0
		MCWBR	8.9	8.1	7.3	5.7	7.1	6.4	5.9	6.3	6.0	4.4	6.5	7.8
Clear-Sky Solar Irradiance	taub	0.204	0.258	0.307	0.331	0.349	0.363	0.390	0.369	0.320	0.276	0.198	0.210	
	taud	1.886	2.042	2.055	2.105	2.252	2.438	2.363	2.392	2.489	2.231	1.920	1.897	
	Ebn at Noon	180	571	726	816	845	845	802	781	748	557	176	11	
	Edh at Noon	16	56	94	117	113	97	102	89	64	46	14	3	
All-Sky Solar Radiation	RadAvg	0.06	0.64	2.25	4.36	5.49	5.22	5.22	3.37	1.82	0.62	0.13	0.01	
	RadStd	0.01	0.06	0.15	0.24	0.40	0.38	0.54	0.28	0.15	0.10	0.01	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	+1.54	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page